Altogether they have 105 cds.
Let n stand for the number of cds Dorothy has.
(i) Write an expression for the number of cds Elaine has.
(ii) Write an expression for the number of cds Michael has.
(iii) Write an expression, in terms of n, for the number of cds all three have together.
(iv) Use this expression to form an equation in n.
(v) Solve this equation and write down the number of cds each has.

4. Tom has two brothers, Colin and Doug
Colin is 4 years younger than Tom
Doug is twice as old as Tom.
Let y stand for Tom's age in years
(i) How old is Colin in terms of y
(ii) How old is Doug in terms of y
(iii) If their ages add up to 24 write an equation in y connecting their ages.
(iv) Solve the equation and find out the age of each brother.

5. The following is a number puzzle
"Think of a number and multiply it by 2 then add 12.
The answer is 3 times the starting number".
What number did I start with?
(i) Which of the following equations corresponds to this number puzzle?

(a) $12n + 2 = 3n$ (b) $2n + 12 = 3$ (c) $2n + 12 = 3n$

(ii) Use the correct equation to solve the puzzle.
(iii) Check your solution works in the puzzle.

6. When 6 is added to a number, the result is 11.
Let x be the number.
(i) Write an equation for this sentence.
(ii) Solve the equation and check that your answer is correct.

7. When 4 is subtracted from twice a number, the result is 12.
Let x be the number.
(i) Write an equation for this sentence.
(ii) Solve the equation and check that your answer is correct.

8. When 4 times a number is increased by 3, the answer is 23.
Let x be the number.
(i) Write an equation for this sentence.
(ii) Solve the equation and check that your answer is correct.

9. When 6 is taken from five times a number, the answer is 14.
Find the number.

10. If I multiply a number by 3 and then subtract 4, the result is the same as twice the number. Find the number.

11. When 3 is taken from five times a number, the result is the same as adding 6 to twice the number. Find the number.

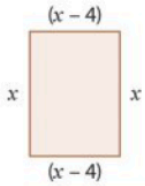
12. One number is 5 greater than another number.
(i) If we let x be the first number write an expression the second number.
(ii) If the two numbers are added together the result is 59. Find the numbers.

13. One number exceeds another by 6.
(i) If we let x be the first number write an expression for the second number.
(ii) When twice the larger number is added to the larger one, the result is 27.
Find the two numbers.

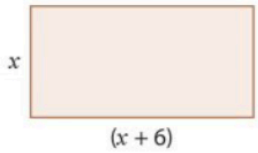
14. Ann is 5 years older than Jean.
If their ages are added the result is 51, find the age of each.

15. Sarah weighs x kg. Her mother weighs $2x$ kg.
Her father weighs 21 kg more than her mother.
(i) Write an expression in terms of x for the weight of her father.
(ii) If the sum of all their weights is 171 kg, form an equation and solve for x.

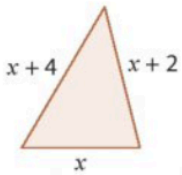
16. One side of a rectangle is 4 metres shorter than the other side as shown in diagram.
If the perimeter of the rectangle measures 24 metres, find the length and the width of the rectangle.



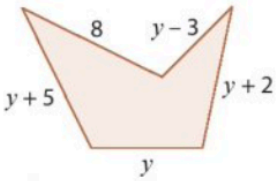
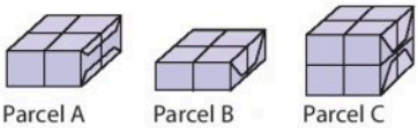
17. One side of a rectangle is 6 m longer than the other side.
(i) If the perimeter of the rectangle is 52 m use the diagram to form an equation in x.
(ii) Solve for x.
(iii) Find the length and the width and hence the area of the rectangle.



18. One side of a triangle is 4 m longer than the base.
The other side is 2 m longer than the base.
(i) If the perimeter of the triangle is 18 m, use the diagram to form an equation in terms of x.
(ii) Solve the equation and write down the length of the three sides.



19. The perimeter of this shape is 32 m.
(i) Find the value of y.
(ii) Find the lengths of all five sides.



20. Parcel A weighs x grams.
Parcel B is 50 grams lighter than parcel A.
Parcel C is three times as heavy as parcel A.
(i) Write an expression for the weight of parcel B in terms of x
(ii) Write an expression for the weight of parcel C in terms of x
(iii) If the total weight of all three parcels is 840 grams, write an equation in terms of x and solve it to find the weight of parcel A
(iv) What is the weight of the two heaviest parcels?